Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

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1 Introduction

The team's definition of a hazard is an "unsafe" condition that may lead to an unintended event that causes an undesirable outcome [Ippolito and Wallace, 1995]. In relation to software development, hazards can be understood as the underlying design decisions that result in potential harm. These risks become apparent when the hazards are not addressed prior to development. These hazards can include technical failures, privacy/security, usability and accessibility risks.

The project's goal is to develop a tool that provides guidance for a physiotherapist's prescribed home exercises. The user's form is analyzed and prompts for adjustments are given. While this can increase the rate and efficacy of recovery, it introduces several hazards. See the Section 3 in the Problem Statement document for more on project goals. The purpose of this document is to identify those hazards, outline assumptions and propose solutions to limit their presence and their frequency.

2 Scope and Purpose of Hazard Analysis

This document outlines the hazards that may occur while operating the tool. The key areas for scope include:

- Technical Hazards: Inability to process user footage, incorrect identification of exercise from different planes, or inaccurate data comparisons, which can result in performance issues and impact the functionality of the tool. Furthermore, these hazards can result in misleading feedback, which could contribute to improper and unsafe exercise techniques.
- Usability Hazards: Usability ensures that the tool has the potential for successfully being adopted. All of the inputs to the system, such as the recording of the exercise, are generated from the user. A user-friendly interface guarantees ease of use, which encourages utilization of this tool. On the other hand, incorrect, on-screen prompts may cause confusion while the user performs the exercise. This may reduce usability and lead to ambiguous guidelines.
- Privacy/Security Hazards: Unnecessary data collection and storage increases the potential of unauthorized access and misuse of personal information. Such situations, such as data breaches, may lead to catastrophic outcomes, which will result in lower adoption rate.

The purpose of this document is to identify risks and their associated impact to determine ways to mitigate them and identify any future occurrences

3 System Boundaries and Components

To divide the system boundaries we outline the key functions of the application.

- Authentication module
  - The user will log in to the application to access prior user data.
- Video streaming module
  - The user will video stream themselves performing a physio exercise pertaining to the knee joint.
- Analysis module
  - The input of the user performing an exercise will be assessed against the requirements and guidelines defined in the backend.

In addition we will consider the following within the hazard analysis:

- A smart device with a camera
  - To allow users to record themselves during the motion and to receive feedback on the exercise performed.
- A database where user data will be stored

These system boundaries outline what is covered within this hazard analysis.

4 Critical Assumptions

- Assume that the user only had mobility issues pertaining to the knee area and with single joint activity.
  - Moreover, it is assumed that the user will perform an exercise that resembles the exercise.
- Assume that only one user is within the frame of the smart device.
- Assume that the user is receptive to feedback.
- Assume that the user knows how to use the smart device and generally how to navigate it when human-centered interfaces are used.
- Assume that the user has a stable connection to the internet.
- Assume that the user is undergoing physiotherapy and was advised to perform the exercise at home.
- Assume that the device used meets the minimum requirements of our system to function.

5 Failure Mode and Effect Analysis

Table 2: Failure Modes and Effects Analysis

Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Authenticate user	Unauthorized access to different patient data	Breach in privacy to personal information	A. Having a weak password	A. User are required to have to create certain password requirements to create an account and sign in.	A. FR1.4 B. SR1 C. NFR 2.3	H1-1
			B. No form of 2FA to protect personal data	B. Users are required to enter an email when creating account for 2FA		
			C. Application keeps user signed in constantly	C. The system will log the user out after 10 minute of inactivity		

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Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Authenticate user	Unable to access user profile	User unable to log into app	A. Technical failure (eg database access down) B. Too many failed login attempts	A. Implement pre-release verification of external library and database connectivity B. Implement a change password option by sending email	A. SR2 B. SR3	H1-2
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	App is unable to provide feedback	A. The environment around the user interferes with the system's detection B. App is unable to use the camera	A. Prompt the user to change their environment to a controlled environment without obstructions to the camera B. Close other apps that may be using camera	A. SR4 B/D. SR5 C. SR6 E. FR2.5	H2-1

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Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
			C. Device with insufficient charge	C. Prompt the user that they need sufficient charge to proceed with the analysis		
			D. App does not have camera access permission	D. Grant permission for app to access camera		
			E. Application does not have sufficient data to analyze user	E. Application support retry attempts for the user's form if not aligned or has obstructions		
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	Application is unable to provide accurate feedback	A. The environment around the user interferes with the system's detection	A. Refer to H2-1A	A. SR4 B. SR2	H2-2

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Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
			B. The system places markers incorrectly	B. Refer to H1-2A		
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	Limited, incorrect or no footage provided for processing different body parts	A. The environment around the user interferes with the system's detection B. Device with insufficient charge C. Application does not have sufficient data to analyze user	A. Refer to H2-1A B. Refer to H2-1C C. Refer to H2-1E	A. SR4 B. SR6 C. FR2.5	H2-3

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Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	May mislead the user	A. Incorrect placement of the body during exercise	A. Provide some sort of outline on the screen where the user has to position themselves to make each view standardized	A. SR7 B. SR4 C. SR8	H3-1
			B. Environment is cluttered and there are interferences during analysis	B. Refer to H2-1C		
			C. Application does not have sufficient data to analyze user	C. System supports retry attempts if insufficient data		
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	Causes feedback to be incorrect and may satisfy the wrong requirements	A. Incorrect placement of the body during exercise	A. Refer to H3-1A	A. SR7 B. SR4	H3-2

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
			B. Environment is cluttered and there are interferences during analysis	B. Refer to H2-1A		
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	App is unable to provide feedback	A. Incorrect placement of the body during exercise B. Environment is cluttered and there are interferences during analysis C. Application does not have sufficient data to analyze user	Refer to H3-1	A. SR7 B. SR4 C. SR8	H3-3

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Real-time feedback	Feedback is slow and does not provide it to the user well	Users cannot correct form and delay in verification of the new form may cause poor performance of the movement	A. Implementation of metrics maybe delaying how prompts are generated	Review how the metrics are being generated and when prompts are being given to the user (i.e. running average but it takes many data points)	A. NFR1.3 B. SR8	H4-1
			B. Application does not have sufficient data to analyze user	Refer to H3-3C		
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	System provides the users with incorrect feedback outside of their physiotherapist's recommendations	System does not know the limitations set by the physiotherapist	Allow a manual input to set daily/weekly limit on increases in-load	A/B. SR9	H4-2

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Design Function	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
				B. Allow a manual input to set daily/weekly limit on increases on mobility		
Analyzing exercise form (monitoring)	Confusing prompts (jargon)	User may not be able to perform the exercise as intended	Prompts are confusing and use jargon in their instructions	A. The system provides a tutorial of the movement during feedback B. Refer to H4-2 C. Design for accessibility (audio cues and visual cues)	A. SR10 B. SR9 C. NFR3.1	H4-3
Analyzing exercise form (monitoring)	Confusing prompts (jargon)	User may harm themselves as a result of the technical jargon	Prompts are confusing and use jargon in their instructions	Refer to H4-3	A. SR10 B. SR9 C. NFR3.1	H4-4

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Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Analyzing exercise form (monitoring)	Confusing prompts (jargon)	User might not know what to correct and become frustrated thus not use the app	Prompts are confusing and use jargon in their instructions	Refer to H4-3	A. SR10 B. SR9 C. NFR3.1	H4-5
Data Management (database)	The system pulls incorrect patient data	This can lead to a security breach and improper access of other user data	Prompts are confusing and use jargon in their instructions	A. The system provides a tutorial of the movement during feedback	A. B. SR9 C. NFR3.1	H5-1
Data Management (database)	The system pulls incorrect patient data	This can affect how feedback is given and stored for future	System incorrectly takes from the wrong userID	Refer to H1-2A	A. SR4	H5-2
Data Management (database)	The system pulls incorrect patient data	There are duplicate entries	Refer to H5-1A	Refer to H5-1A	A. NFR2.1	H5-3

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Design Function	**Failure Modes**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Data Management (database)	Unable to access user profile	System is unable to log in user	Refer to H1-2	Refer to H1-2	A. SR2 B. SR3	H5-4

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

References

Laura M. Ippolito and Dolores R. Wallace. *A Study on Hazard Analysis in High Integrity Software Standards and Guidelines*. 1995. URL https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir5589.pdf. U.S. Department of Commerce, Technology Administration, National Institute of Standards and Technology.